Unit 1: Quiz - Set 1

Topics: Introduction to Data Structures; Array Representation and Operations; Pointers
Format: 10 MCQ Questions | **Time:** 5 minutes

Q1. What is the primary advantage of using an array over a linked list?

a. Dynamic size adjustments
b. Constant time access to any element given its index
c. Faster insertion and deletion at the middle
d. Uses less memory overhead for large datasets

Correct Answer: b

Q2. The memory address of the first element of an array is commonly referred to as the:

a. Peak address
b. Base address
c. First pointer
d. Null address

Correct Answer: b

Q3. Based on the 2D matrix representation below, how are elements uniquely identified?

0,0	0,1	0,2
1,0	1,1	1,2

Representation of Matrix

a. By a single sequential index
b. By a memory address only
c. By a combination of a row index and a column index
d. By the value stored inside the cell

Correct Answer: c

Q4. In C programming, what does the & operator do when placed before a variable?

a. Returns the value of the variable
b. Dereferences the variable
c. Returns the memory address of the variable
d. Casts the variable to a pointer

Correct Answer: c

Q5. Which of the following data structures is considered non-linear?

a. Queue
b. Stack
c. Linked List
d. Graph

Correct Answer: d

Q6. What usually happens if you try to access an array element outside its bounds in C?

a. The compiler throws a syntax error
b. The program automatically resizes the array
c. It results in undefined behavior or a segmentation fault
d. It returns a standard NULL value

Correct Answer: c

Q7. A pointer that has not been initialized to any valid memory address and is currently pointing to a random location is known as a:

a. Null pointer
b. Dangling pointer
c. Wild pointer
d. Void pointer

Correct Answer: c

Q8. How is a 2D array typically laid out in contiguous memory in languages like C?

Row vs Column Major Order

a. Diagonal-major order
b. Column-major order
c. Random memory allocation
d. Row-major order

Correct Answer: d

Q9. Which of the following operations on an unsorted array has a worst-case time complexity of O(n)?

a. Finding the maximum value
b. Accessing the 5th element
c. Updating the first element
d. Finding the memory size of the array

Correct Answer: a

Q10. If ptr is a pointer to an integer (and assuming an integer takes 4 bytes), what does ptr++ do?

a. Increments the value ptr points to by 1
b. Advances the pointer's address by 1 byte
c. Advances the pointer's address by 4 bytes (to the next integer)
d. Creates a new pointer

Correct Answer: c